

CIS 568 · DATA VISUALIZATION · FINAL PROJECT

NTSB Aviation Accidents

Project README & Usage Guide

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Overview

The final project is an interactive, real-time data visualization for researching US aviation accidents from 1962 through 2025. NTSB reports from foreign countries were excluded from the database; the scope is limited to US civil aviation.

GitHub Pages: michaeljevan.github.io/CIS_568_Final_Project

Project Video: youtu.be/zmMK5TGLTpi

Running Locally

The recommended workflow is to open the project folder in VS Code and launch it with the *Go Live* extension.

Required files (must all live in the same directory):

- `index.html`
- `USonly_v3.csv`
- `topojson-client.min.js`
- `Counties-10m.json`
- `states-10m.json`

Project Video

The final project video is a 15-minute .mp4, not a slide presentation. It demonstrates the full functionality of the visualization and includes a use-case scenario near the end.

The video focuses on *what the app does*, not on how it was built. A few technical choices are mentioned in passing, but for implementation specifics, see the in-app **Documentation** drawer or the companion `Technical_Report.md` in the repository.

Interaction Guide

Map Controls

Click any dot or cluster to open the accident popup. Click the NTSB case number in the popup to open the full investigation docket on the NTSB website (information is generally available for incidents from 2009 onward).

CLUSTER

Groups nearby accidents; zoom in to separate them. Large datasets may render slowly.

HEATMAP

Shows accident density weighted by injury severity.

ALL DOTS

Plots every individual accident as a single point.

Filters

- **Single-click** any filter button to toggle it on or off.
- **Double-click** to isolate that category — hides all others in the group.
- **Double-click again** to restore all filters in that group.
- The **year range slider** filters by accident date.
- **Injury-level buttons** in the top legend control dot colors on the map.

Basemap Options

DARK / STREET

Switches between a dark basemap (zoomed out) and a street map (zoomed in).

SECTIONAL

Overlays FAA VFR sectional charts — best between zoom levels 6 and 12. Sectional view automatically switches to **ALL DOTS** mode.

TOPOGRAPHIC

Interactive topographic map from outer zoom levels down to street level.

SPIKE

Interactive US spike map with hover-over statistics. Click any state to open the state/county drill-down view.

Timeline

Animates accidents year by year from 1962 through 2025. Press ► to play and ✕ to close and return to your previous view. Statistics update in real time as the timeline advances, and the timeline works at any zoom level.

Dataset

168,793 US civil aviation accidents covering 1962–2025, sourced from the NTSB Aviation Accident Database.

- Some older records have only approximate city-level coordinates — these are flagged with a ! indicator in the popup.
- Pre-2009 investigation dockets may not be publicly released on the NTSB website.

Built With

- **Leaflet** — interactive map with cluster, heatmap, and individual dot rendering modes.
- **MarkerCluster** — groups nearby markers into clickable clusters that split apart on zoom.
- **D3.js** — line chart, bar chart, donut chart, sparklines, US state choropleth, and spike map.
- **TopoJSON** — US state and county boundary geometry for the choropleth and drill-down maps.
- **PapaParse** — parses the 168k-row CSV dataset client-side in the browser.
- **noUiSlider** — dual-handle year range slider.

All data is loaded from a static CSV file; no server or database is required. Basemap tiles are served by CARTO (dark/voyager), the FAA (sectional), and OpenTopoMap.

Screenshots

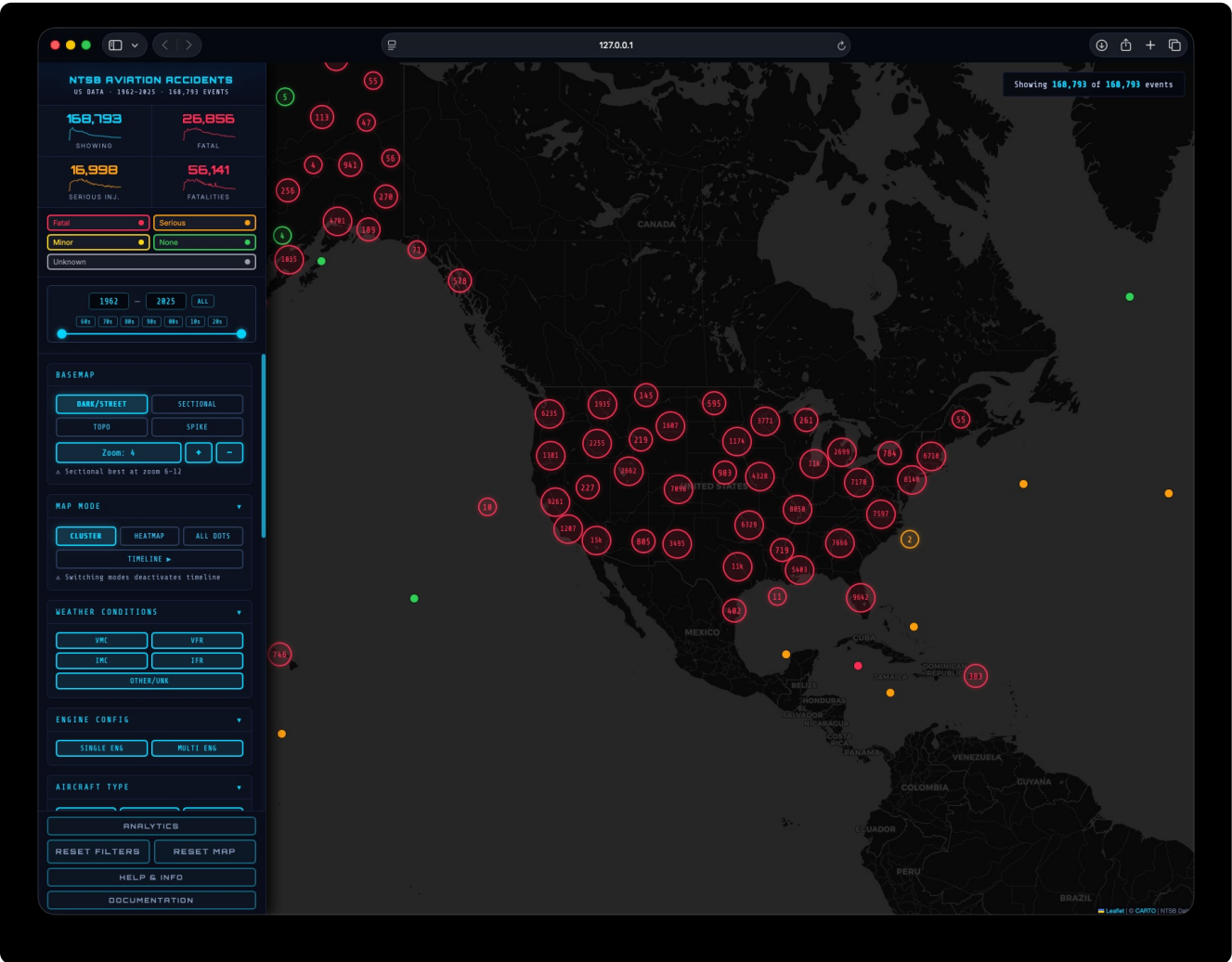


FIGURE 1 • CLUSTER MODE (DARK BASEMAP)

Full-dataset view with accident counts grouped into clusters. Cluster color reflects the worst injury severity in the group. Sidebar shows live summary statistics and the full filter stack.

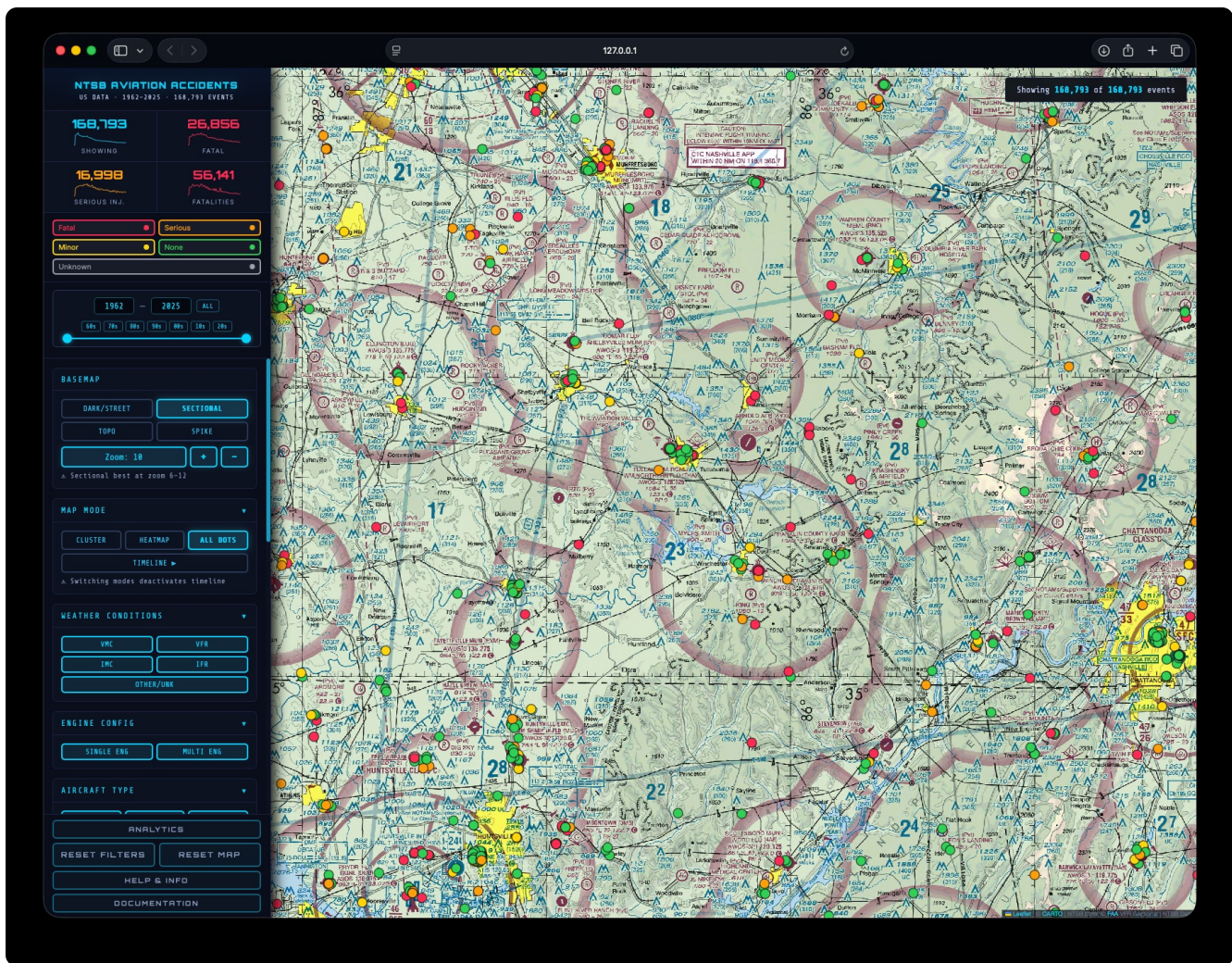


FIGURE 2 · FAA VFR SECTIONAL OVERLAY

Accidents plotted over an FAA VFR sectional chart. Automatically switches to **All Dots** mode; works best between zoom levels 6 and 12. Dot color encodes injury severity.

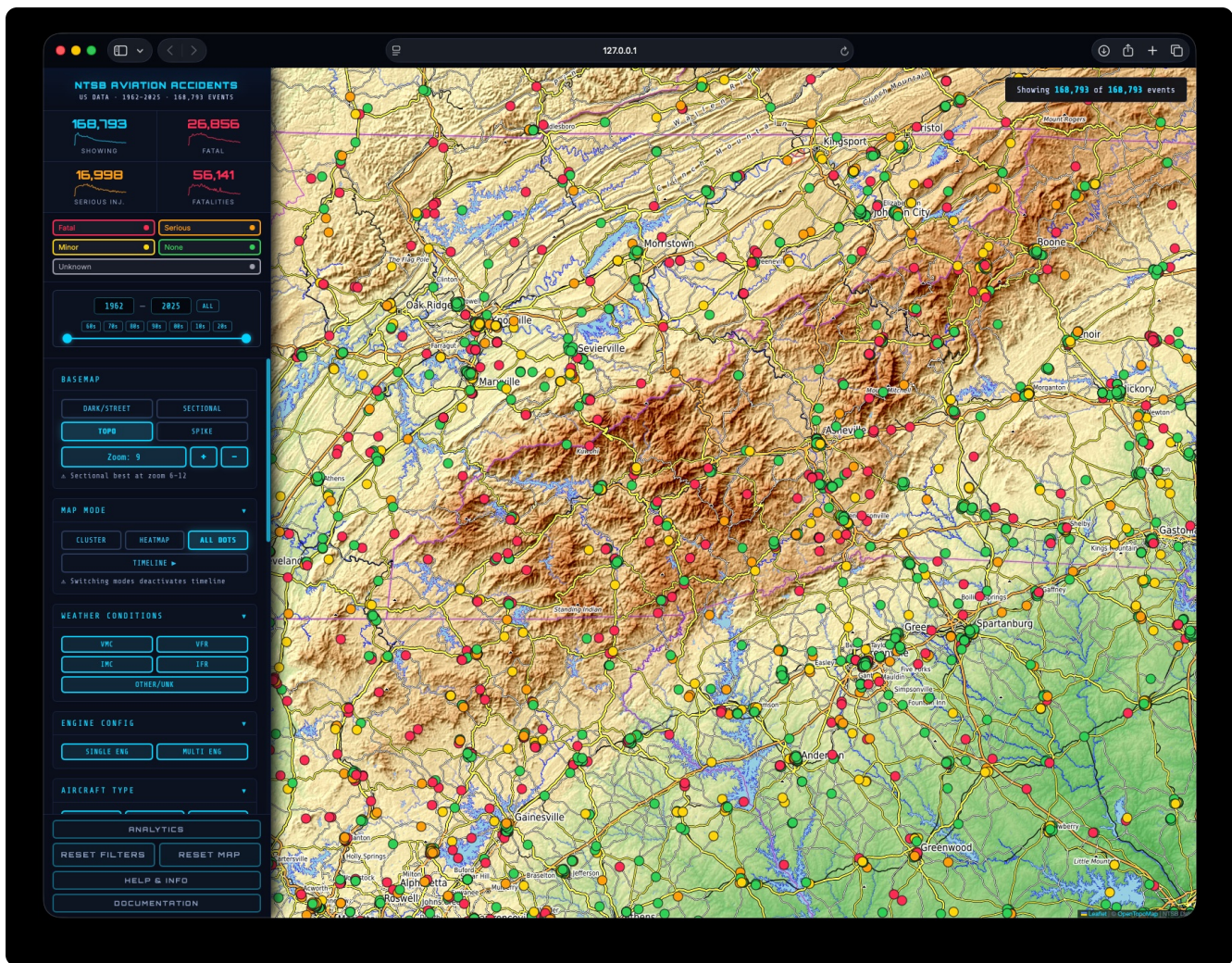


FIGURE 3 · TOPOGRAPHIC BASEMAP

Accident dots overlaid on OpenTopoMap terrain tiles, useful for correlating accident density with mountainous terrain and other geographic features.



FIGURE 4 · US SPIKE MAP

Vertical spikes scale proportionally to accident count per state. Hover for full statistics; click any state to open the county drill-down view.

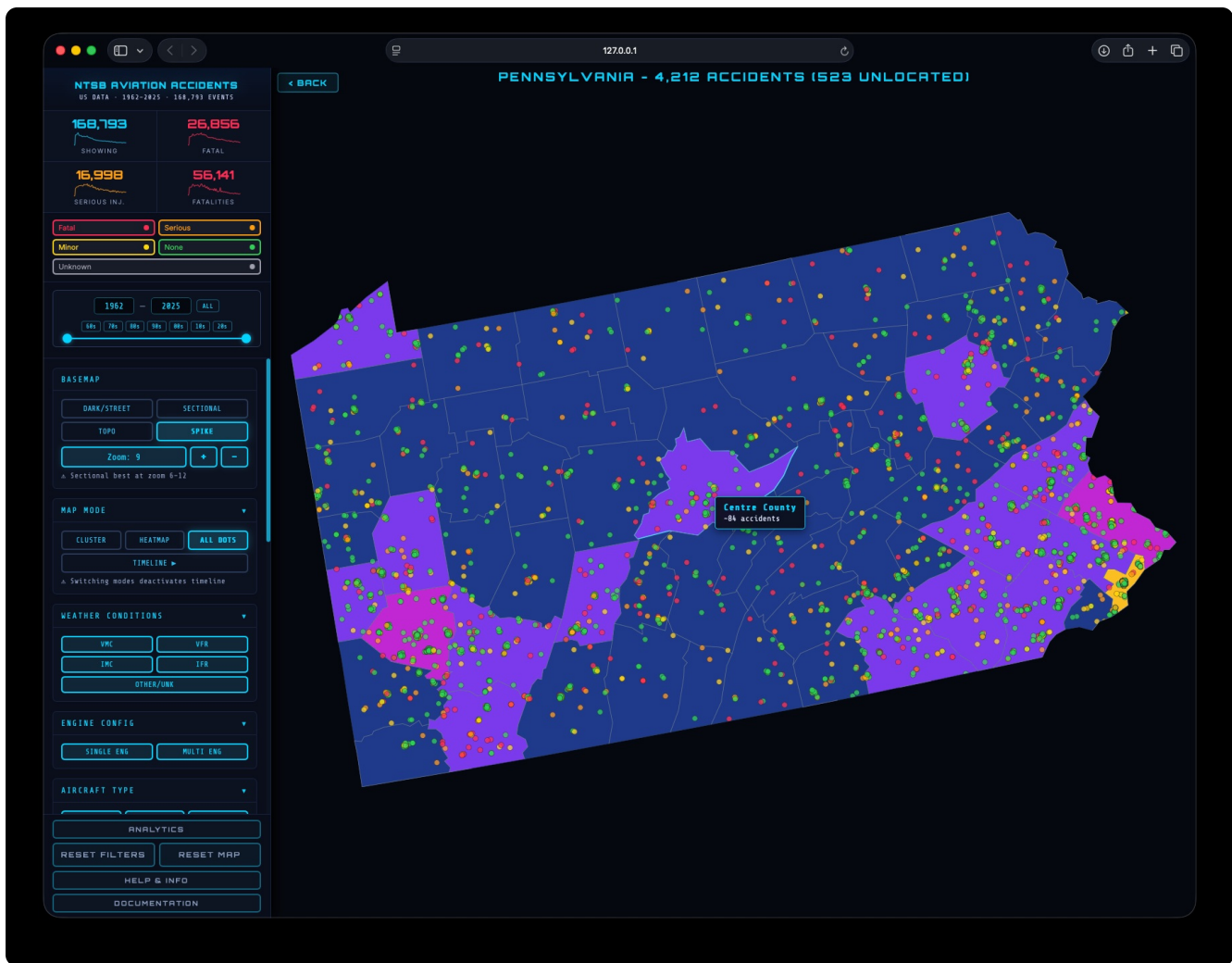


FIGURE 5 • COUNTY DRILL-DOWN (PENNSYLVANIA)

County-level choropleth colored by accident density, with individual accident dots rendered on a canvas overlay. Hover tooltips display county counts; the tilde prefix indicates approximate minimums when unlocated records are present.

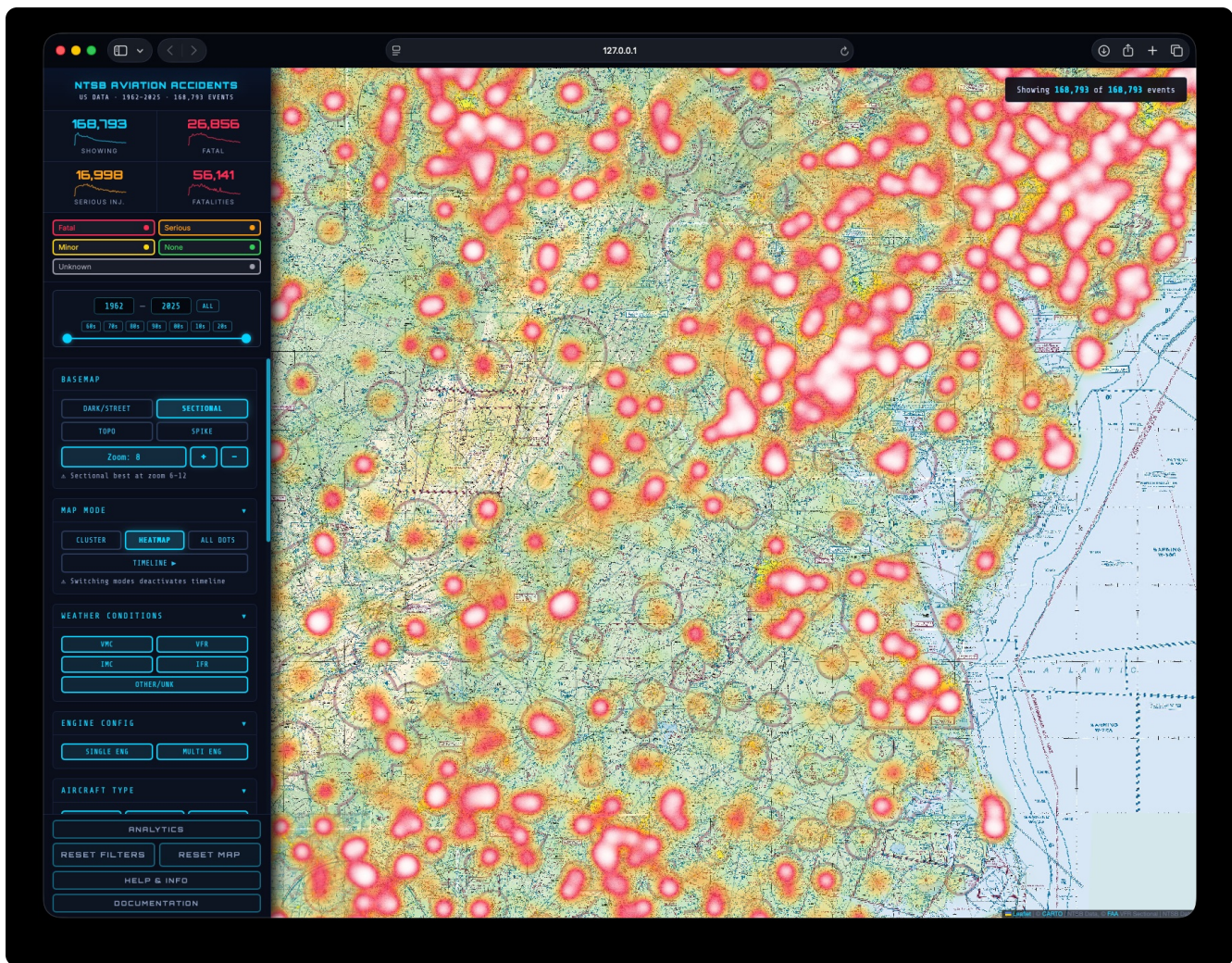


FIGURE 6 • HEATMAP MODE (OVER SECTIONAL)

Accident density rendered as a continuous color gradient, weighted by injury severity so fatal events contribute more visual weight than minor incidents.

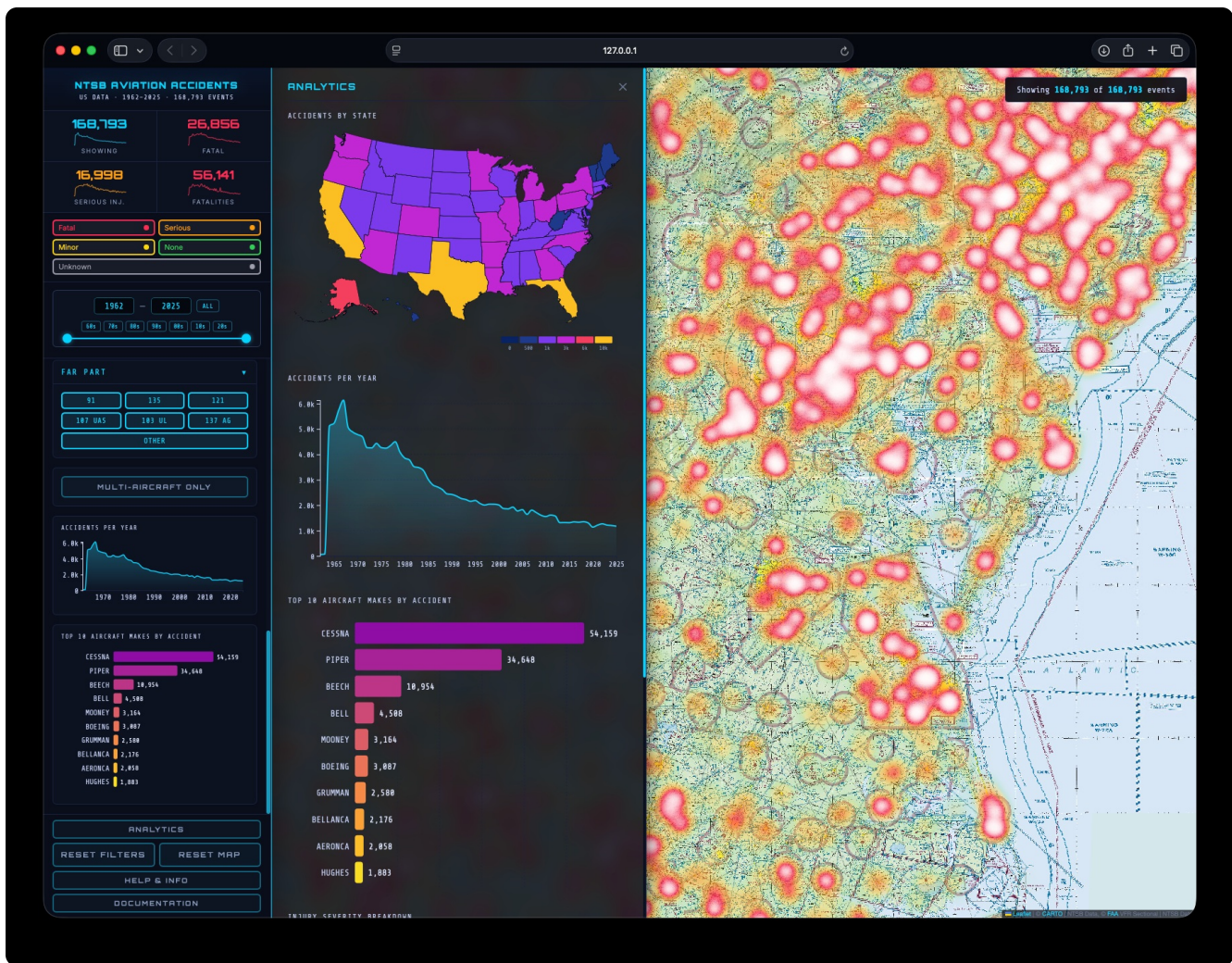


FIGURE 7 · ANALYTICS DRAWER

Slide-out analytics panel showing the US state choropleth, accidents-per-year trend, and top aircraft makes by accident count. All charts update in real time as filters change.